

IPVS

2026-06-11

Packages

```
library(arrow)
```

```
## Warning: package 'arrow' was built under R version 4.5.3
```

```
##  
## Attaching package: 'arrow'
```

```
## The following object is masked from 'package:utils':  
##  
##     timestamp
```

```
library(readr)
```

```
## Warning: package 'readr' was built under R version 4.5.3
```

```
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.5.3
```

```
##  
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':  
##  
##     filter, lag
```

```
## The following objects are masked from 'package:base':  
##  
##     intersect, setdiff, setequal, union
```

```
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.5.3
```

```
library(tidyr)
library(purrr)
```

```
## Warning: package 'purrr' was built under R version 4.5.3
```

```
library(knitr)
library(kableExtra)
```

```
## Warning: package 'kableExtra' was built under R version 4.5.3
```

```
##
## Attaching package: 'kableExtra'
```

```
## The following object is masked from 'package:dplyr':
##
##   group_rows
```

Dataset

```
ipvs <- read_parquet("C:/Users/alyss/Desktop/Brasil/Analysis/dados_current - 2026-06-09/Dados/resultados/contextual/ipvs_municipios_sp.parquet")
```

Missingness

```
sum(is.na(ipvs))
```

```
## [1] 0
```

Clean

```
ipvs <- ipvs |>
  mutate(
    ipvs_low =
      ipvs_pct_grupo1 + ipvs_pct_grupo2,
    ipvs_medium =
      ipvs_pct_grupo3 + ipvs_pct_grupo4,
    ipvs_high =
      ipvs_pct_grupo5 + ipvs_pct_grupo6)
```

Distribution of High IPVS

```
ipvs_summary <- ipvs %>%
  select(ipvs_high) %>%
  rename(value = ipvs_high) %>%
  mutate(
    Quartile = ntile(desc(value), 4)
  ) %>%
  group_by(Quartile) %>%
  summarise(
    n = n(),
    pct = round(100 * n() / nrow(ipvs), 1),
    Median = median(value, na.rm = TRUE),
    Min = min(value, na.rm = TRUE),
    Max = max(value, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  mutate(
    Variable = "IPVS High",
    Range = paste0(round(Min, 2), "-", round(Max, 2))
  ) %>%
  select(Variable, Quartile, n, pct, Median, Range)
```

ipvs_summary

```
## # A tibble: 4 × 6
##   Variable Quartile      n  pct Median Range
##   <chr>      <int> <int> <dbl> <dbl> <chr>
## 1 IPVS High      1  162  25.1  35.6 25.32-84.06
## 2 IPVS High      2  161   25   18.8 13.67-25.28
## 3 IPVS High      3  161   25   10.3  6.57-13.67
## 4 IPVS High      4  161   25    2.51 0-6.56
```

Neat Table

```
ipvs_summary %>%
  kable(
    digits = 1,
    caption = "Distribution of Municipalities by High/Very High IPVS",
    col.names = c(
      "High IPVS",
      "Quartile",
      "n",
      "%",
      "Median (%)",
      "Range"
    )
  ) %>%
  collapse_rows(columns = 1) %>%
  kable_styling(full_width = FALSE)
```

Distribution of Municipalities by High/Very High IPVS

	High IPVS	Quartile	n	%	Median (%)	Range
IPVS High		1	162	25.1	35.6	25.32-84.06
		2	161	25.0	18.8	13.67-25.28
		3	161	25.0	10.3	6.57-13.67
		4	161	25.0	2.5	0-6.56

Normality

```
ipvs_vars <- c("ipvs_low", "ipvs_medium", "ipvs_high")

ipvs %>%
  select(all_of(ipvs_vars)) %>%
  pivot_longer(
    cols = everything(),
    names_to = "ipvs",
    values_to = "value"
  ) %>%
  ggplot(aes(x = value)) +
  geom_histogram(bins = 20) +
  facet_wrap(~ ipvs, scales = "free") +
  theme_minimal()
```

